

Deploy mission-critical private 5G for airport operations with stringent SLAs.

Transportation and Logistics × Tactical and mission-critical communications × Private 5G / LTE · OS137 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
73 is the world moving	54 can we play, can we win	HIGH 1 named in the evidence	€2.4bn partial evidence · per year	NOW budgeted procurement within 1...	L1 11 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

We deploy mission-critical private 5G networks for airport operations, engineered to stringent SLAs for availability, reliability, latency, and security. This opportunity targets airport operators and ground service providers who need deterministic, secure wireless for operational and safety-critical communications. The need is urgent because airports face rising drone incursions and must modernize aging critical communications infrastructure.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€3.8bn €831m – €22.2bn	€2.4bn €518m – €13.8bn	€28.4m €6.2m – €166m	partial
Observed: tendered contract value, annualised	€115m	€115m	€1.4m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Transportation and Logistics (EU27_2020)	113,199 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Private 5G / LTE	32.5 % of enterprises (10+ employees)	proxy	Eurostat · isoc_eb_iotn2 · E_IOT1	2021
Annual value of one engagement	€792k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-26..2026-08-05
Size-weighted buyer base	14,748 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. Internet of Things by NACE Rev. 2 activity series E_IOT1 is used as a proxy for Private 5G / LTE: No private-network series; industrial IoT use as proxy. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders. 1 of 1 declared geographies are outside the European reference data (GB); the estimate covers the European footprint only.

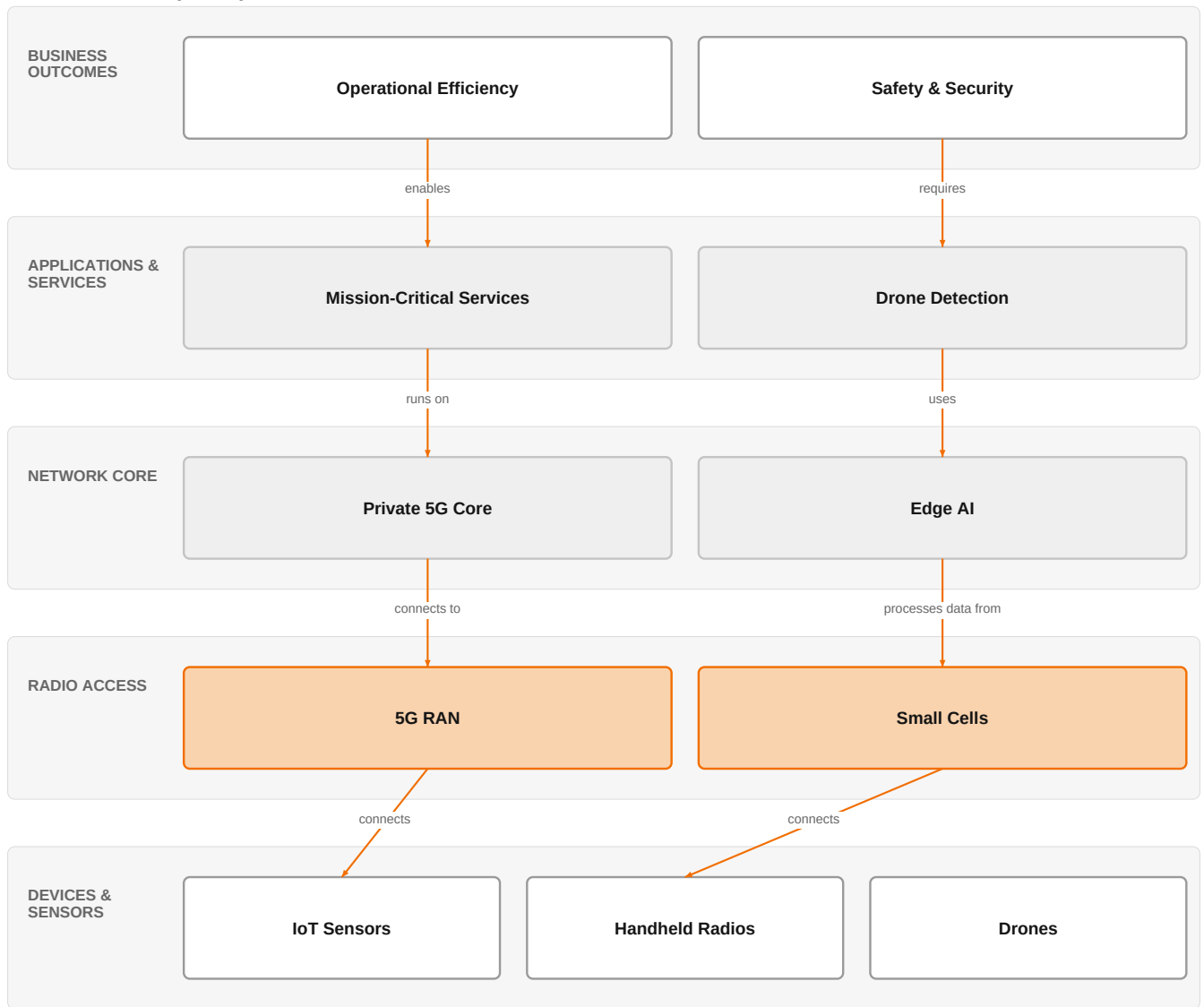
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€115m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-26..2026-08-05
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 8 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size. 1 of 1 declared geographies are outside the European reference data (GB); the estimate covers the European footprint only.

The solution, in one picture

Private 5G for Airport Operations



■ Orange asset ■ Partner ■ Customer-owned ■ Third party / to be sourced

This diagram shows how a private 5G network underpins mission-critical airport operations, from devices to business outcomes.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Mission-critical services in 4G/5G are now standardized, meeting stringent requirements for availability, reliability, latency, security, and QoS, making them viable for critical infrastructure like airports. Private LTE/5G networks offer greater determinism and reliability than Wi-Fi, but also introduce a unique threat landscape that must be addressed. Airports are actively investing in critical communications infrastructure, as evidenced by Heathrow's requirement for operational support and maintenance of mission-critical radio and cellular systems. Additionally, drone incursions at airports are a rising issue, with notable incidents reported worldwide, driving the need for robust detection and response capabilities.

Sources: SIG-C0BAE4D32A34 openalex.org 2025-08-19; SIG-6FBA261C5571 International Journal 2025-08-08; SIG-01D38D7333B8 contractsfinder.servic 2026-08-12; SIG-FDBC54B7D40 Vietnam.vn 2026-08-12; SIG-9FCAF3E022B5 ASIS International 2026-06-30

Who buys, and why now

The buyer is typically the Chief Technology Officer or Head of Airport Operations, who signs for network infrastructure that supports safety-critical services. The pain is felt by operations teams who rely on voice, video, and data for ground handling, security, and emergency response; they experience dropped calls, latency, and coverage gaps that hinder coordination. The trigger is often a major incident—such as a drone incursion or a system failure—that exposes the

limitations of existing Wi-Fi or public networks, prompting a budgeted procurement within the next year.

Why it is hot now — every claim bound to a dated source

- Heathrow Airport requires operational support and maintenance for mission-critical radio and cellular systems. [SIG-01D38D7333B8]
- Mission-critical services in 4G/5G are standardized for public safety and critical infrastructure. [SIG-C0BAE4D32A34]
- SITA is hiring a Senior Specialist for Mission Critical Radio & Cellular Systems at Heathrow Airport, indicating investment in critical communications infrastructure. [SIG-01D38D7333B8]
- Mission-critical services standardized in 4G/5G meet stringent requirements for availability, reliability, latency, security, and QoS, essential for critical infrastructure domains. [SIG-C0BAE4D32A34]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would deliver a private mobile network (L1) tailored to airport operations, leveraging our position as a Leader in the Gartner Magic Quadrant for 4G and 5G Private Mobile Network Services. The engagement begins with a site survey and spectrum assessment, followed by network design and deployment using our partner ecosystem (Samsung, Huawei) for RAN and core components. We then integrate mission-critical services (MCPTT, MCVideo, MCDATA) and provide ongoing managed services with stringent SLAs, drawing on our IoT, Digital, Defense and security experts, and Orange Group researchers for continuous optimization.

Why Orange can win

Orange can win because we are a Leader in the Gartner Magic Quadrant for 4G and 5G Private Mobile Network Services, and we bring deep expertise from our IoT, Digital, and Defense and security teams. Our reference with Point One Navigation demonstrates our ability to deliver high-precision, mission-critical solutions. However, the competitive intensity is high, and our differentiation must be proven through our SLA commitment and sovereign, secure infrastructure.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Private mobile networks	offer	L1	offer provides the technology in a matching domain	unconfirmed
Samsung	partner	L2	partner provides the required technology	unconfirmed
Huawei	partner	L2	partner provides the required technology	unconfirmed
Gartner Magic Quadrant 2026 — 4G and 5G Private Mobile Network Services (Leader)	analyst position	SUP	analyst recognition covers this technology or use case	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Digital experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Point One Navigation	reference	SUP	published reference in this vertical, different use case	unconfirmed

Competitive landscape

HIGH competitive intensity (score 9.0; 1 of 8 listed players appear in this topic's own evidence)

Crowded, with heavyweight incumbents already visible in the evidence. Win on a specific differentiator or do not bid.

Competitor	Category	Position here	Basis	Evidence
Verizon Business	Telecom operator peer	sells Private 5G / LTE; competes in Connectivity Solutions	evidenced	SIG-35ECED6AF5 BB
BT Business	Telecom operator peer	sells Private 5G / LTE; competes in Connectivity Solutions	structural	—

Deutsche Telekom / T-Systems	Telecom operator peer	sells Private 5G / LTE; competes in Connectivity Solutions	structural	—
NTT DATA	Global systems integrator	sells Private 5G / LTE; competes in Connectivity Solutions	structural	—
Proximus	Telecom operator peer	sells Private 5G / LTE; competes in Connectivity Solutions	structural	—
Swisscom	Telecom operator peer	sells Private 5G / LTE; competes in Connectivity Solutions	structural	—
Telefonica Tech	Telecom operator peer	sells Private 5G / LTE; competes in Connectivity Solutions	structural	—
Telia Company	Telecom operator peer	sells Private 5G / LTE; competes in Connectivity Solutions	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 What are your current pain points with your existing critical communications systems, and how do they impact daily operations?
- 2 Have you experienced any recent incidents (e.g., drone incursions, communication failures) that highlighted the need for more reliable connectivity?
- 3 What are your specific requirements for availability, reliability, latency, and security in your mission-critical communications?
- 4 Do you have a budget allocated for upgrading your communications infrastructure within the next 12 months?
- 5 Who is the primary decision-maker for network infrastructure investments, and what is their main concern?
- 6 How do you currently handle drone detection and response, and what gaps exist in your current approach?

Objections you will meet

They will say	You can answer
We already have Wi-Fi and public 5G; why do we need a private network?	Wi-Fi and public 5G can't guarantee the determinism and reliability required for mission-critical services. Private LTE/5G offers dedicated resources, lower latency, and higher availability, which are essential for safety and operational efficiency.
Private 5G is too expensive and complex to deploy.	While the initial investment is significant, the total cost of ownership can be lower than maintaining multiple legacy systems. Our managed service model reduces complexity, and we can phase the deployment to align with your budget.
We're concerned about the security of private networks.	Security is a top priority. We design our private networks with robust security architecture, and we have defense and security experts who ensure compliance with the highest standards. We also provide continuous monitoring and threat mitigation.

Risks and unknowns

The main risk is that airports may be slow to adopt private 5G due to high upfront costs and the complexity of integrating with legacy systems. There is also uncertainty about spectrum availability and regulatory approval in different jurisdictions. Additionally, the security of private LTE/5G networks is a concern, and we must demonstrate our ability to mitigate evolving threats. Finally, the competitive landscape is crowded, and we may face price pressure from telecom peers.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a European airport operator's mission-critical private 5G tender, highlighting Orange's Leader position in the Gartner Magic Quadrant and the capabilities of our IoT, digital, and defense and security experts, and referencing Point One Navigation as a proof point.
Sales	In a meeting with the head of operations at a major European airport, say: 'Orange is a Leader in the Gartner Magic Quadrant for private mobile network services, and we have the expertise to deliver mission-critical private 5G with the stringent SLAs your operations require. Let's discuss how we can support your critical communications.'

Strategist / Innovator	Commission a deep-dive brief on the private 5G opportunity for airport operations, leveraging the Gartner Magic Quadrant Leader position and Orange Group researchers to assess the competitive landscape and identify the key differentiators for the aviation sector.
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Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-01D38D7333B8	2026-08-12	contractsfinder.servi ce.g...	1	Resource Requirement - Senior Specialist/Engineer Service Operations for Mission Critical Radio...
SIG-5EF8BAA50BF5	2026-07-14	Iconic Research and Engin...	1	5G Core SA Enablement for Private Industrial Networks: Opportunities for Oil, Gas, Mining, Logi...
SIG-C0BAE4D32A34	2025-08-19	openalex.org	1	Mission-Critical Services in 4G/5G and Beyond: Standardization, Key Challenges, and Future Pers...
SIG-6FBA261C5571	2025-08-08	International Journal on ...	1	Security Challenges and Solutions in Private Wireless (LTE/5G) Network Implementations
SIG-CC7D9567F38D	2025-07-31	International Journal for...	1	Private 5G Network: Security, Challenges, and Comparison with Wi-Fi
SIG-9E3ACF1CE0DC	2025-07-16	Emitor: Jurnal Teknik Ele...	1	Implementation of 5G Standalone Private Network for UHD Video Streaming
SIG-DD814ADD17F5	2025-01-01	cordis.europa.eu	1	Enabling Autonomous Aerial Systems through Operational Methods and Technological Enhancements
SIG-35ECED6AF5BB	2026-08-12	bing.com	2	Lockheed Martin converts Verizon 5G into subscription drone alerts for critical infrastructure
SIG-4A3CF0B146B7	2026-08-12	bing.com	2	Private 5G Network Deployment Tracker & Forecasts 2026-2030 Covers 9,300 Engagements Across 1...
SIG-FDBC854B7D40	2026-08-12	Vietnam.vn	2	5 incursions notables de drones dans des aeroports du monde entier. - Vietnam.vn
SIG-9FCAF3E022B5	2026-06-30	ASIS International	2	Drone Reports Near U.S. Airports Reflect Rising Issue for Critical Infrastructure - ASIS Intern...

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive_landscape (named entities that were never supplied: Heathrow Airport, SITA, Lockheed Martin); diagram (2 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:14+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:17+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-18T20:48:01+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.